

CLOUD COMPUTING LAB

LAB O1

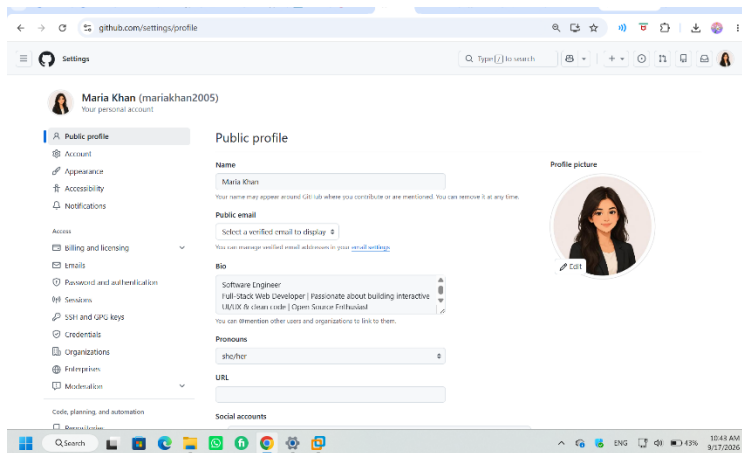
GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

=

NAME	MARIA KHAN
REGISTRATION NUMBER	2024-BSE-41
COURSE CODE & SECTION	CLOUD COMPUTING (SE-665) SECTION A
GITHUB USERNAME	@Mariakhan2005

Task 1: GitHub Account Setup & Portal Registration

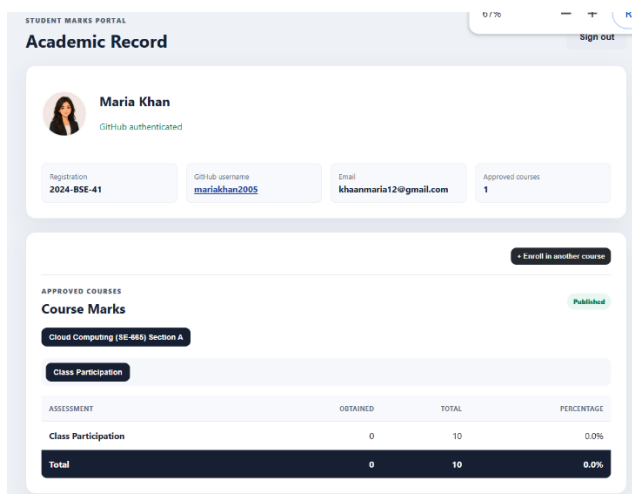
GitHub profile



Portal login page

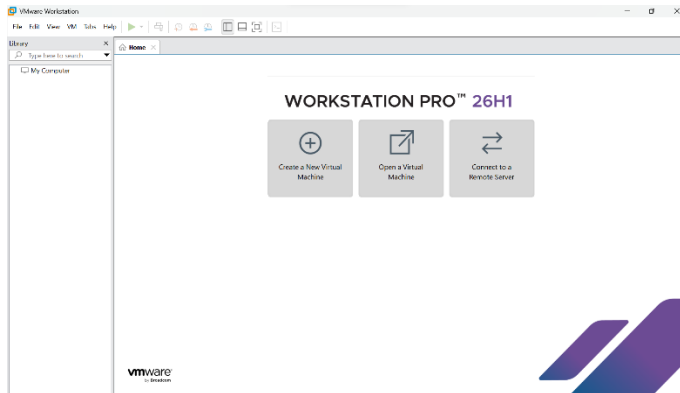


Portal enrollment submitted



Task 2 - 7: VMware & VM Creation Setup

VMware WorkStation



VM creation wizard



Vm typical configuration

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☐ Store virtual disk as a single file
☒ Split virtual disk into multiple files
 Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

[Help](#)
[< Back](#)
[Next >](#)
[Cancel](#)

Vm ubuntu is selected

☒ Installer disc image file (iso):
 [Browse...](#)
☐ Ubuntu 64-bit Server 24.04.5 detected.
 To use Easy Install, insert the first disc of the set.

Vm configuration summary

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

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[Help](#)
[< Back](#)
[Next >](#)
[Cancel](#)

Ubuntu installer_boot

Home x Ubuntu 64-bit x

GNU GRUB version 2.12

Try or install Ubuntu Server
 Ubuntu Server with the HWE kernel
 Test memory

Use the ↑ and ↓ keys to select which entry is highlighted. Press enter to boot the selected OS, 'e' to edit the commands before booting or 'c' for a command-line. The highlighted entry will be executed automatically in 10s.

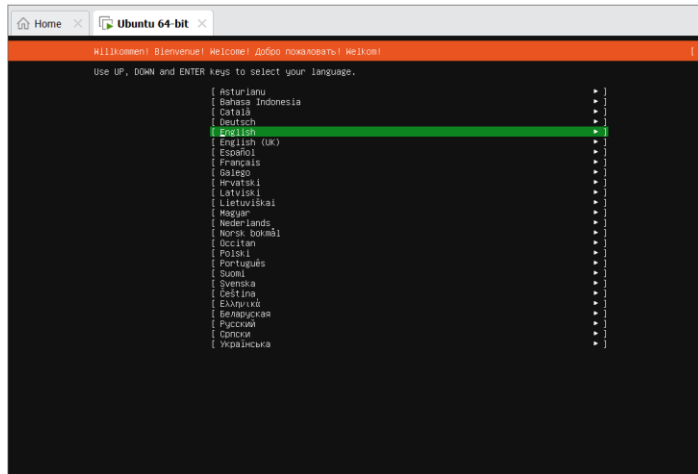
Install Ubuntu 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".

[I Finished Installing](#)
[Help](#)

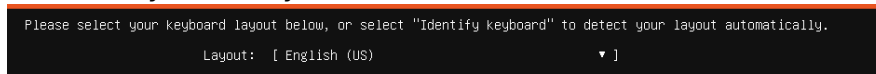
or press Ctrl+G.

Task 8: Language, Keyboard & Network Setup

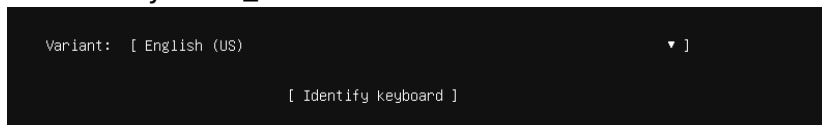
Ubuntu language



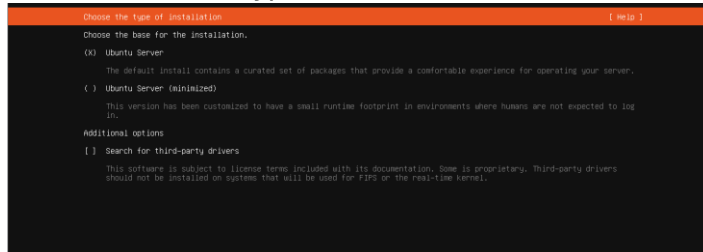
Ubuntu keyboard_layout



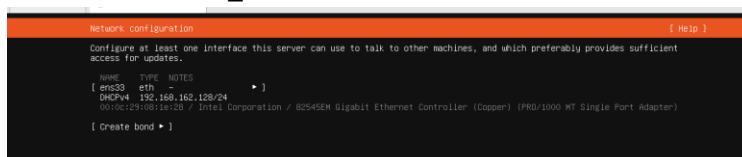
Ubuntu keyboard_variant



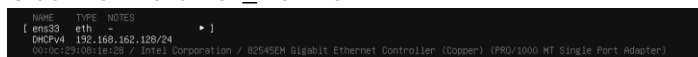
Ubuntu installer_type



Ubuntu network_interface

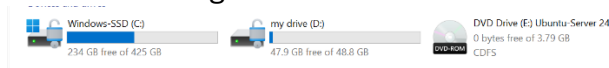


Ubuntu installer_network



Task 9 - 12: Storage, Partitions & Warnings

Available storage



Ubuntu guided_storage

guided storage configuration [Help]

Configure a guided storage layout, or create a custom one:

Use an entire disk

[/dev/sda local disk 20.000G ▾]

Set up this disk as an LVM group

Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

Also create a recovery key.
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

Custom storage layout

Ubuntu selected_disk

USED DEVICES			
DEVICE	TYPE	SIZE	
[ubuntu-vg (new)	LVM volume group	18.222G	▸]
ubuntu-lv	new, to be formatted as ext4, mounted at /	10.000G	▸]
[/dev/sda	local disk	20.000G	▸]
partition 1	new, BIOS grub spacer	1.000M	▸]
partition 2	new, to be formatted as ext4, mounted at /boot	1.771G	▸]
partition 3	new, PV of LVM volume group ubuntu-vg	18.225G	▸]

Ubuntu disk_details

Storage configuration [Help]

FILE SYSTEM SUMMARY

MOUNT POINT	SIZE	TYPE	DEVICE TYPE
[/	10.000G	new ext4	new LVM logical volume ▸]
[/boot	1.771G	new ext4	new partition of local disk ▸]

AVAILABLE DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	18.222G ▸]
free space		8.222G ▸]

Create software RAID (md) ▸]

Create volume group (LVM) ▸]

USED DEVICES

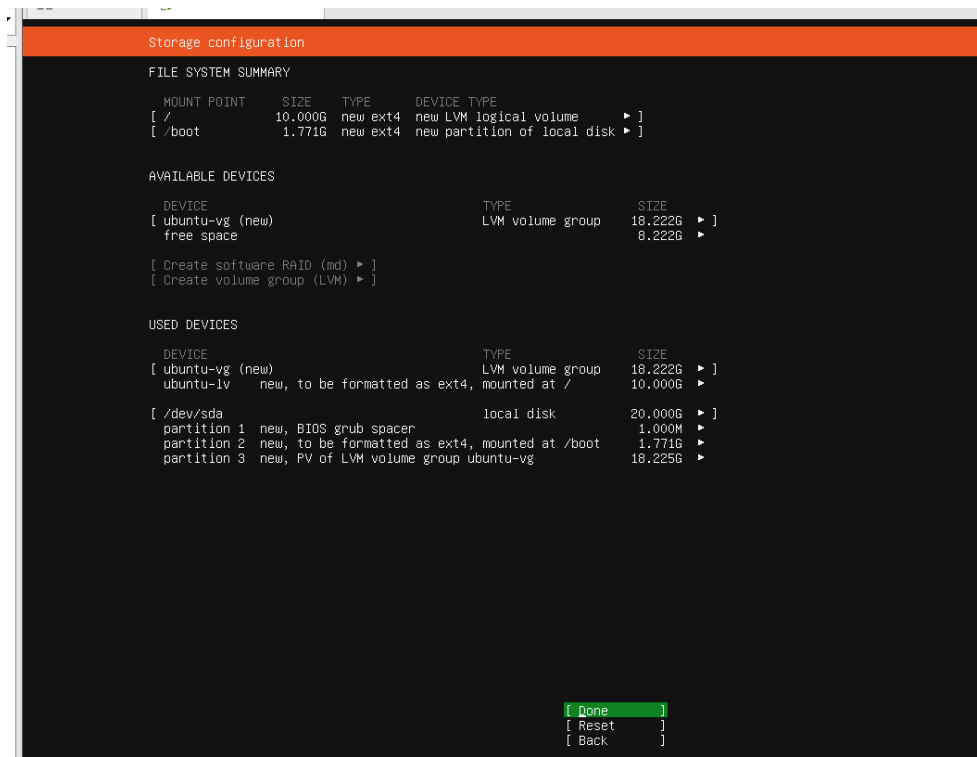
DEVICE	TYPE	SIZE
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partition 1	new, BIOS grub spacer	1.000M ▸]
partition 2	new, to be formatted as ext4, mounted at /boot	1.771G ▸]
partition 3	new, PV of LVM volume group ubuntu-vg	18.225G ▸]

Done

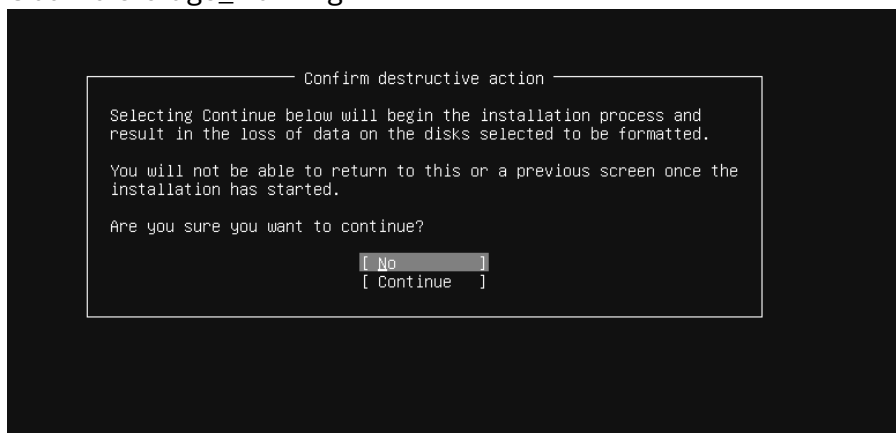
Reset

Back

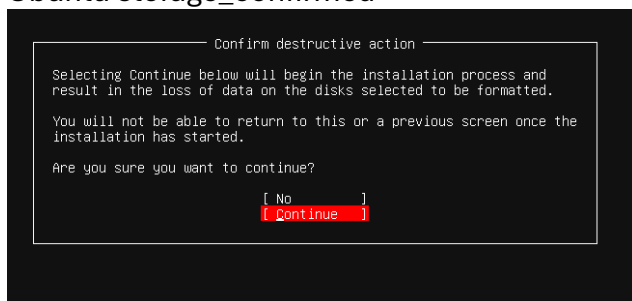
Ubuntu partition_layout



Ubuntu storage_warning



Ubuntu storage_confirmed



Task 13: Profile Setup (Identity Verification)

Ubuntu actual_name

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Maria Khan_

Ubuntu server_name

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Maria Khan

Your servers name: Ubuntu
The name it uses when it talks to other computers.

Ubuntu username

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Maria Khan

Your servers name: Ubuntu
The name it uses when it talks to other computers.

Pick a username: mariakhan2005_

Task 14 - 15: Installation, Reboot & Login Terminal

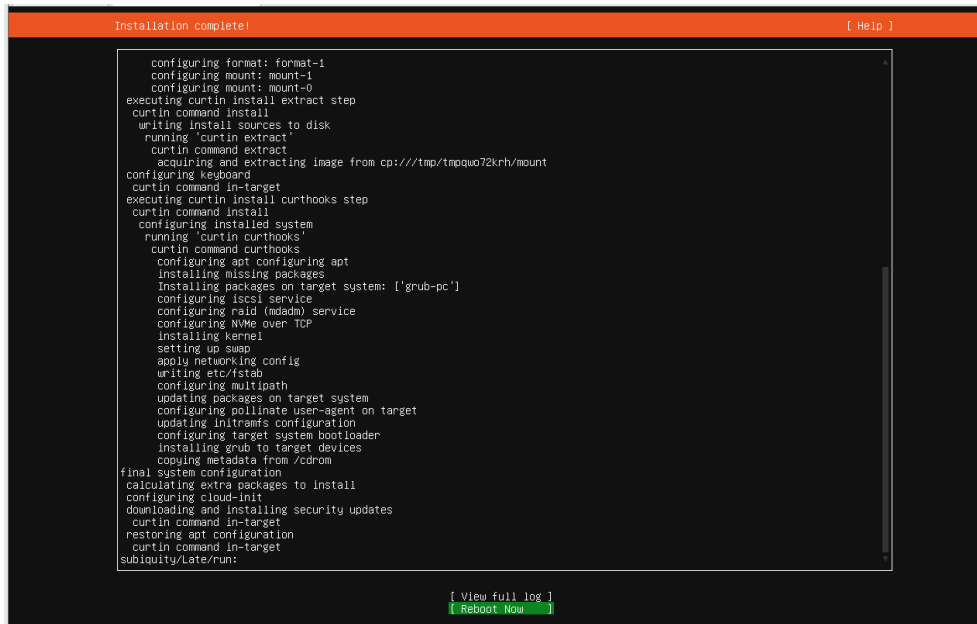
Ubuntu installation_progress

Updating system [Help]

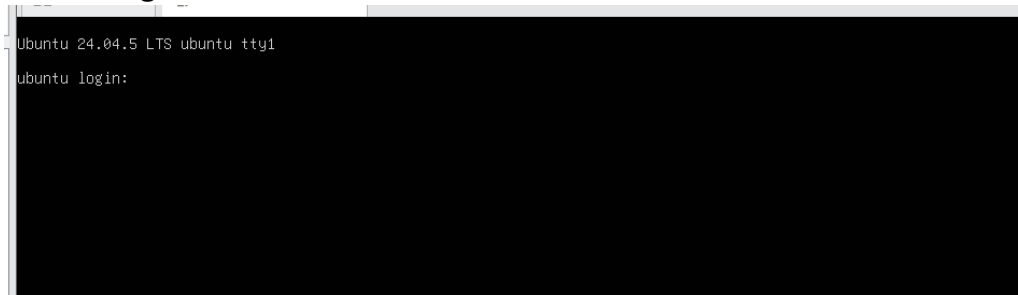
```
configuring partition: partition-2
configuring format: format-0
configuring lvm_voigroup: lvm_voigroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpqwo72krh/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
Installing packages on target system: ['grub-pc']
configuring iscsi service
configuring raid (mdadm) service
configuring NvMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
updating initramfs configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target /
```

[View full log]

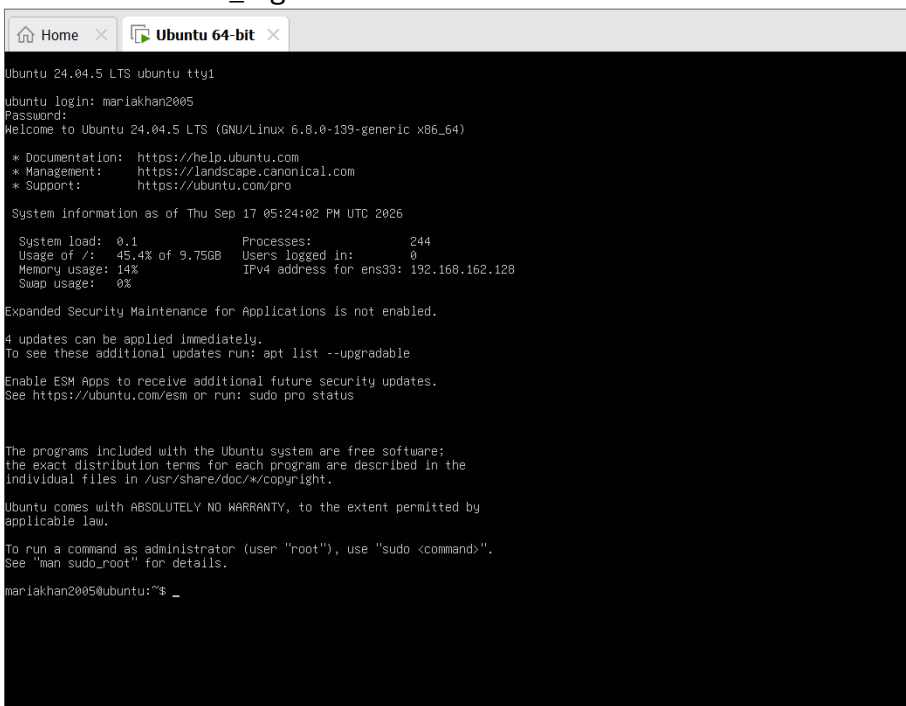
Ubuntu installation_complete



Ubuntu login_screen



Ubuntu terminal_login



Ubuntu identity_verified

See man page for details:

```
mariakhan2005@ubuntu:~$ _
```

Task 16 - 17: IP Address & Remote Windows SSH

Ubuntu ip_addr_command

```
mariakhan2005@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:08:1e:28 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.162.128/24 metric 100 brd 192.168.162.255 scope global dynamic ens33
        valid_lft 1611sec preferred_lft 1611sec
    inet6 fe80::20c:29ff:fe08:1e28/64 scope link
        valid_lft forever preferred_lft forever
mariakhan2005@ubuntu:~$ _
```

Ubuntu ip_address

```
altname enp2s1
inet 192.168.162.128/24 metric 100 brd 192.168.162.255 scope global dynamic ens33
    valid_lft 1611sec preferred_lft 1611sec
inet6 fe80::20c:29ff:fe08:1e28/64 scope link
    valid_lft forever preferred_lft forever
mariakhan2005@ubuntu:~$ _
```

Windows ssh_command

```
C:\Windows\System32>ssh mariakhan2005@192.168.162.128
The authenticity of host '192.168.162.128 (192.168.162.128)' can't be established.
ED25519 key fingerprint is SHA256:rFVNd790Y1hHHz5wq4mIL95+Kc/LHqetxs1VY0Bvmko.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.162.128' (ED25519) to the list of known hosts.
mariakhan2005@192.168.162.128's password:
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-139-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Fri Sep 18 01:11:32 PM UTC 2026

System load:	0.01	Processes:	221
Usage of /:	46.0% of 9.75GB	Users logged in:	1
Memory usage:	16%	IPv4 address for ens33:	192.168.162.128
Swap usage:	0%		

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

```
mariakhan2005@ubuntu:~$
```

Windows ssh_fingerprint

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.162.128' (ED25519) to the list of known hosts.
```

Windows ssh_login

```
mariakhan2005@192.168.162.128's password:
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-139-generic x86_64)

* Documentation:  https://help.ubuntu.com
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

mariakhan2005@ubuntu:~$
```

Windows ssh identity

```
mariakhan2005@ubuntu:~$ _
```

Final PDF Generation solution_title_page.png

CLOUD COMPUTING LAB

LAB 01

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

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